

Avaye Dawadi

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TECHNICAL EXPERIENCE

PrepByAI

06 2021 — Present

ACT Test Prep Website at prepbyai.com

- Launched and maintained a service for free and adaptive ACT test prep that serves 400+ users with 2000+ questions.
- Developed the website using React and Flask and deployed to an AWS EC2 instance.
- Integrated an adaptive algorithm that scales question difficulty to the ability of the user.
- Supervised continued operation and maintenance of the website.

Georgia Tech Research Institute (GTRI) - Information Communications Laboratory

06 2021 — 07 2021

Student Intern

Atlanta, Georgia

- Worked alone with a mentor to do R&D into web components in the Lit library while using proper UI/UX and testing technique.
- Presented research to ~100 GTRI staff and faculty at the end of the 6-week internship.
- Published two components to NPM each with 100+ downloads: [Idle Timeout Component](#), [Pugh Chart Component](#)

Georgia Tech Research Institute (GTRI) - Georgia Tech Robotarium

06 2022 — Present

Student Intern

Atlanta, Georgia

- Collaborating in a team of 3 to translate Robotarium Python into block code to improve overall Robotarium access.
- Utilizing Flask and Docker to deploy a simulator backend for use with the block code.
- Facilitating translation of Robotarium concepts (kinematics, dynamics, position error, formation control, etc.) for individuals not well versed in robotics.

VEX Robotics

08 2018 — 05 2022

Team Leader and Lead Programmer ([Robotics Git Repo](#))

Chamblee, Georgia

- Mentored a team of 7 in proper programming and building practices.
- Programmed the robot in C++ (PROS library from Purdue University) and achieved a top 200 skills score in the world mid-season.
- Implemented and tuned a PID algorithm to assist in self-correction for autonomous robot function.
- Succeeded in building and programming a Georgia state finalist (2nd place) robot that also qualified for the worlds competition in Dallas (roughly top 15% of teams qualify).

ISEF Finalist

07 2020 — 05 2021

International Science and Engineering Fair

- Implemented the A* Pathfinding algorithm to find shortest paths on floor plans: [ISEF Project](#)
- Researched Euclidean distance vs. Manhattan distance, priority queues, the Von Neumann neighborhood, and sets vs. arrays to create the most efficient algorithm possible. Better implementations resulted in ~430% improvement in runtime.
- Implemented algorithm in Python and hosted end product on a Flask website.
- Applied a Convolutional Neural Network on TensorFlow to classify floor plans with ~98% accuracy.
- Achieved 1st place at the Dekalb County Science Fair and qualified for ISEF.

SKILLS

Programming Languages	Python, JavaScript, C++, Java
Frameworks/Libraries	React, TensorFlow, NodeJS, Electron, Flask
Other Tools	Git, Linux, AWS EC2, Google Optimize, Storybook, Puppeteer
Communication	English (Native fluency), German (Intermediate level)

EDUCATION

Chamblee Charter High School, Graduate with Honors and Student of Distinction

Georgia Institute of Technology, Computer Science (Expected Class of 2024)

ACTIVITIES/AWARDS

- Team member (current) in a Deep Learning Playground application for the Data Science club at Georgia Tech
- 2nd Place at Technology Student Association (TSA) National Competition for building an image anomaly detection program as an Electron App over large image data sets.
- M3 math modeling competition honorable mention (\$ 200 scholarship). Top 5% of teams.
- Former software developer intern at myPalate (health food startup).
- Former Secretary of Communication in the Interact Club (subsidiary of Rotary Club)